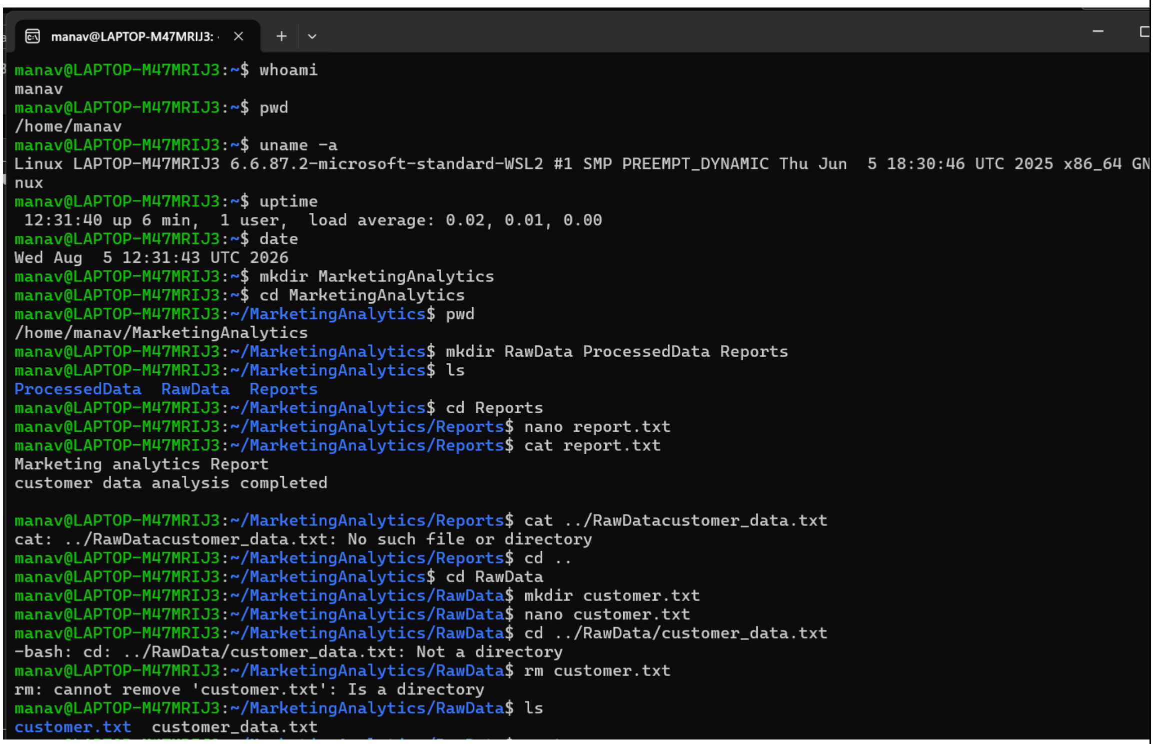
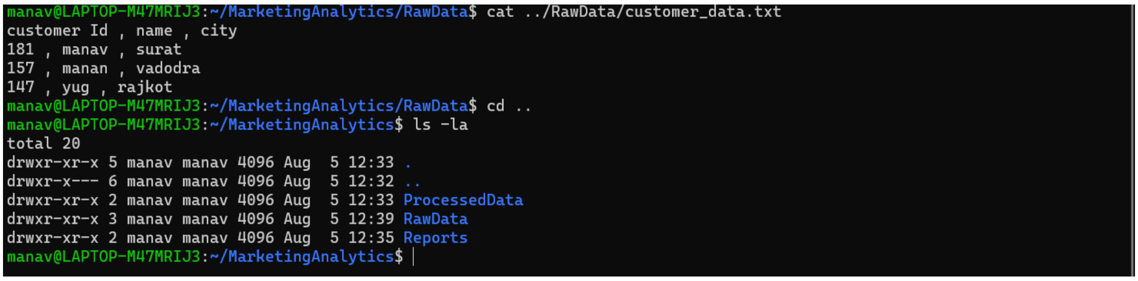
Practical 1	**AIM:** Linux System Administration and File Management for Marketing Analytics Team
Problem Definition	**Scenario:** You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data, and creating backups for the Marketing Analytics team. **Tasks:** 1. Identify the current user. 2. Find the current working directory. 3. Display system information. 4. Check system uptime and date. 5. Create a directory named MarketingAnalytics. 6. Create subdirectories: RawData, ProcessedData, Reports. 7. Create data and report files. 8. Add sample content. 9. Display file contents. 10. List all files and directories. 11. Search for report files. 12. Search specific content inside files. 13. Copy report files to Backup directory. 14. Rename files. 15. Move files between directories. 16. Delete a file. Delete an empty directory.
Output	



```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ cat ../RawData/customer_data.txt
customer Id , name , city
181 , manav , surat
157 , manan , vadodra
147 , yug , rajkot
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ cd ..
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ ls -la
total 20
drwxr-xr-x 5 manav manav 4096 Aug  5 12:33 .
drwxr-x--- 6 manav manav 4096 Aug  5 12:32 ..
drwxr-xr-x 2 manav manav 4096 Aug  5 12:33 ProcessedData
drwxr-xr-x 3 manav manav 4096 Aug  5 12:39 RawData
drwxr-xr-x 2 manav manav 4096 Aug  5 12:35 Reports
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ |
```

Key Questions / Analysis

Q1. What is the difference between absolute and relative paths?
Ans:- An absolute path specifies the complete location of a file or directory starting from the root (/) of the filesystem, e.g. /home/analyst01/MarketingAnalytics/RawData. It works the same way regardless of the current working directory. A relative path specifies the location relative to the current working directory, e.g. RawData/sales_data.csv when already inside MarketingAnalytics. Relative paths are shorter and convenient for scripts run from a known location, while absolute paths are safer for scripts that may be executed from anywhere.

Q2. Why are Linux permissions important for file security?
Ans:- Linux uses a permission model (read, write, execute) applied separately to the owner, group, and others for every file and directory. This ensures that sensitive marketing data — such as customer records or sales reports — can only be viewed or modified by authorized users. Permissions prevent accidental or malicious deletion/modification of data, support multi-user collaboration (e.g. giving the analytics team read/write access while restricting outsiders), and form the foundation of the overall security model of the operating system.

Q3. What is the purpose of the grep command?
Ans:- 'grep' (Global Regular Expression Print) searches the content of files for lines matching a specified text pattern or regular expression and prints those matching lines. It is essential for quickly locating specific information — such as a keyword, error message, or data value — inside large log files or reports without opening and reading them manually, e.g. searching all report files for the word 'Conversion' to pull out performance figures instantly.

Q4. How does Linux support efficient file management?
Ans:- Linux provides a rich set of lightweight command-line utilities (mkdir, cp, mv, rm, find, grep, tar, etc.) that can be combined and scripted for automation, supports a hierarchical filesystem for logical organisation of data, offers fine-grained permission and ownership controls, allows redirection and piping to chain commands together, and integrates scheduling tools like cron for recurring tasks such as backups. This combination makes it possible to manage very large volumes of project data quickly, reliably, and in an automatable way.

Q5. Why are backups important in enterprise environments?

Ans:- Backups protect an organisation against data loss caused by hardware failure, accidental deletion, software bugs, or cyberattacks such as ransomware. In an enterprise setting like DataAnalytics Inc., losing raw sales data or client reports could halt business operations, cause financial loss, and damage client trust. Regular, automated backups (e.g. via cron jobs) ensure business continuity, support disaster recovery, and help meet compliance and audit requirements.

Supplementary Problems

1. **Create a shell script to automate project directory creation.**

```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ cat project_directory.sh
#!/bin/bash

echo "Enter Project Name:"
read project

mkdir -p $project/{RawData,ProcessedData,Reports,Scripts,Backup}

echo "Project Directory Created Successfully."

tree $project
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ |
```

output

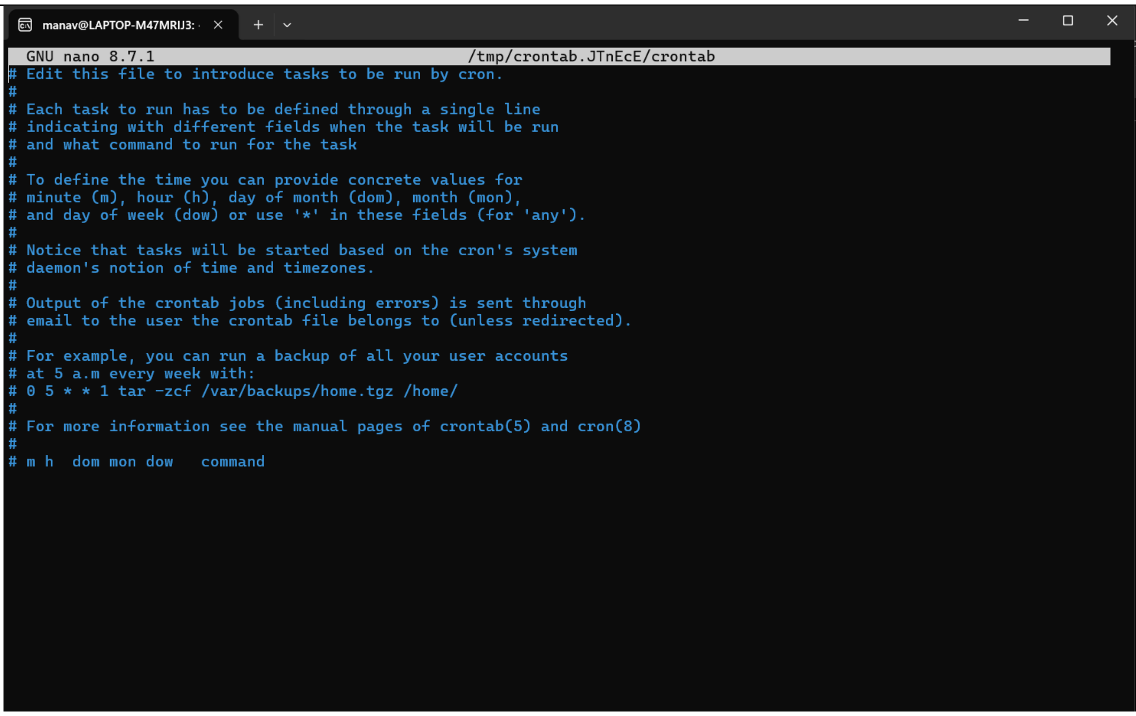
```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ ls
customer.txt  customer_data.txt  project-directory.sh  project_directory.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ vim project_directory.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ chmod +x project_directory.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ ./project_directory.sh
Enter Project Name:
Marketing-Analytics
Project Directory Created Successfully.
./project_directory.sh: line 10: tree: command not found
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics/RawData$ |
```

2. **Find the top 10 largest files in the system.**

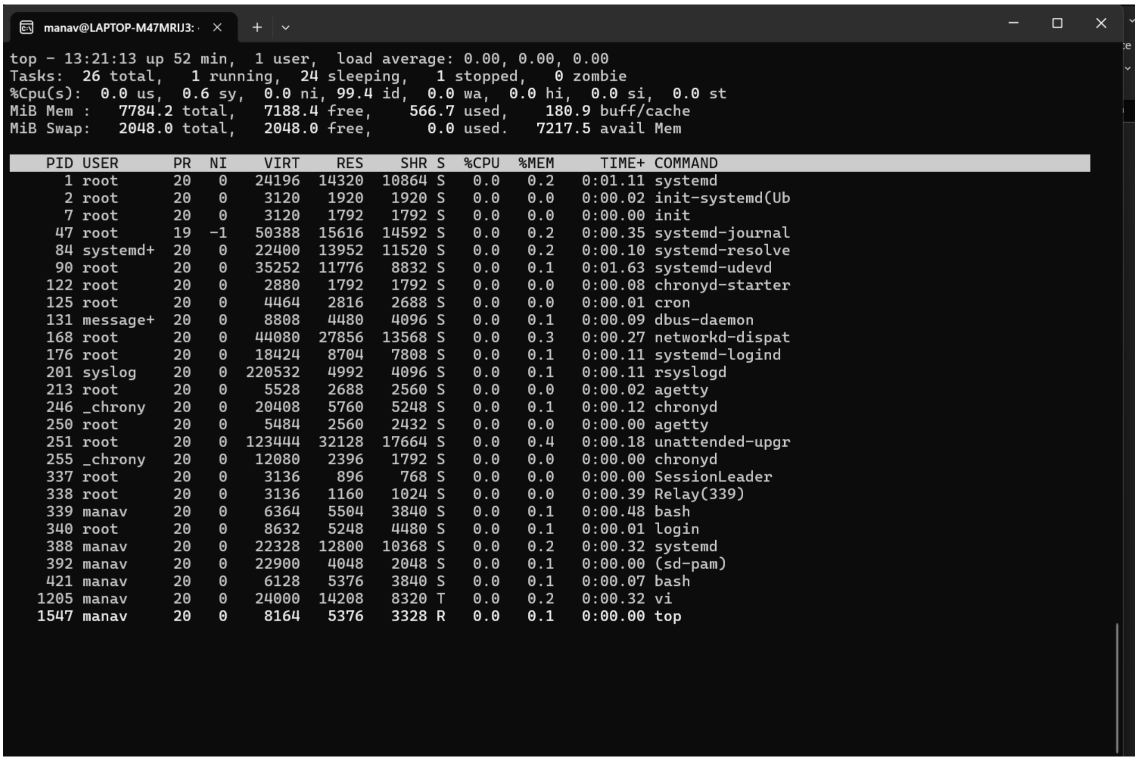
```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ nano largest.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ cat largest.sh
#!/bin/bash

echo "Top 10 Largest Files in the System ---->"

find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ ./largest.sh
-bash: ./largest.sh: Permission denied
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ nano largest.sh
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ ./largest.sh
-bash: ./largest.sh: Permission denied
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ cat largest.sh
echo "Top 10 Largest Files in the System ---->"
find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ |
```

3. **. Schedule a backup task using cron**

```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ crontab -e
no crontab for manav - using an empty one
Select an editor.  To change later, run select-editor again.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny
  4. /bin/ed

Choose 1-4 [1]: 1
No modification made
```



4. Monitor CPU utilization continuously using top/htop.



```
manav@LAPTOP-M47MRIJ3: ·

    0[                    0.0%]   4[                              0.0%]   8[                              0.0%] 12[                    0.0%]
    1[                    0.0%]   5[                              0.0%]   9[                              0.0%] 13[                    0.0%]
    2[                    0.0%]   6[                              0.0%]  10[                              0.0%] 14[                    0.0%]
    3[                    0.0%]   7[                              0.0%]  11[                              0.0%] 15[                    0.0%]
  Mem[|||||||||                                        425M/7.60G] Tasks: 28, 14 thr, 0 kthr; 1 running
  Swp[                                                   0K/2.00G] Load average: 0.07 0.02 0.00
                                                                   Uptime: 00:54:26

  Main  I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
      1 root        20   0 24196 14320 10864 S   0.0  0.2  0:01.18 /sbin/init
      2 root        20   0  3120  1920  1920 S   0.0  0.0  0:00.01 /init
      7 root        20   0  3120  1792  1792 S   0.0  0.0  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8
      8 root        20   0  3120  1792     0 S   0.0  0.0  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8
      9 root        20   0  3120  1920     0 S   0.0  0.0  0:00.00 /init
     47 root        19  -1 50388 15616 14592 S   0.0  0.2  0:00.35 /usr/lib/systemd/systemd-journald
     84 systemd-re  20   0 22400 13952 11520 S   0.0  0.2  0:00.10 /usr/lib/systemd/systemd-resolved
     90 root        20   0 35252 11776  8832 S   0.0  0.1  0:01.64 /usr/lib/systemd/systemd-udevd
    122 root        20   0  2880  1792  1792 S   0.0  0.0  0:00.08 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -
    125 root        20   0  4464  2816  2688 S   0.0  0.0  0:00.01 /usr/sbin/cron -f -P
    131 messagebus  20   0  8808  4480  4096 S   0.0  0.1  0:00.11 @dbus-daemon --system --address=systemd: --nofork --n
    168 root        20   0 44080 27856 13568 S   0.0  0.3  0:00.27 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-s
    176 root        20   0 18424  8704  7808 S   0.0  0.1  0:00.12 /usr/lib/systemd/systemd-logind
    201 syslog      20   0  215M  4992  4096 S   0.0  0.1  0:00.07 /usr/sbin/rsyslogd -n -iNONE
    213 root        20   0  5528  2688  2560 S   0.0  0.0  0:00.02 /usr/sbin/agetty --noreset --noclear --issue-file=/et
    246 _chrony     20   0 20408  5760  5248 S   0.0  0.1  0:00.13 /usr/sbin/chronyd -n -F 1 -x
    247 syslog      20   0  215M  4992     0 S   0.0  0.1  0:00.02 /usr/sbin/rsyslogd -n -iNONE
    248 syslog      20   0  215M  4992     0 S   0.0  0.1  0:00.00 /usr/sbin/rsyslogd -n -iNONE
    249 syslog      20   0  215M  4992     0 S   0.0  0.1  0:00.01 /usr/sbin/rsyslogd -n -iNONE
    250 root        20   0  5484  2560  2432 S   0.0  0.0  0:00.00 /usr/sbin/agetty --noreset --noclear --issue-file=/et
    251 root        20   0  120M 32128 17664 S   0.0  0.4  0:00.18 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
    255 _chrony     20   0 12080  2396  1792 S   0.0  0.0  0:00.00 /usr/sbin/chronyd -n -F 1 -x
    299 root        20   0  120M 32128     0 S   0.0  0.4  0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
    337 root        20   0  3136   896   768 S   0.0  0.0  0:00.00 /init
    338 root        20   0  3136  1160  1024 S   0.0  0.0  0:00.42 /init
    339 manav       20   0  6364  5504  3840 S   0.0  0.1  0:00.51 -bash
    340 root        20   0  8632  5248  4480 S   0.0  0.1  0:00.01 login -- manav
    388 manav       20   0 22328 12800 10368 S   0.0  0.2  0:00.32 /usr/lib/systemd/systemd --user
    392 manav       20   0 22900  4048  2048 S   0.0  0.1  0:00.00 (sd-pam)
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

5. . Generate a disk usage report for all project directories.

```
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ htop
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ du -sh */ > disk_report.txt
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ cat disk_report.txt
4.0K    ProcessedData/
44K     RawData/
8.0K    Reports/
manav@LAPTOP-M47MRIJ3:~/MarketingAnalytics$ |
```